

Multiple Choice

1. **Answer:** B
Options: A) commutative semi group; B) abelian group; C) non-abelian group; D) None of these
Note: Correct option: B - abelian group
2. **Answer:** B
Options: A) 2; B) 3; C) 6; D) 12
Note: Correct option: B - 3
3. **Answer:** C
Options: A) symmetric only; B) anti-symmetric only; C) both symmetric and anti-symmetric; D) an equivalence relation
Note: Correct option: C - both symmetric and anti-symmetric
4. **Answer:** B
Options: A) -5; B) -4; C) -3; D) -2
Note: Correct option: B - -4
5. **Answer:** B
Options: A) 3; B) π ; C) 3π ; D) $3/2$
Note: Correct option: B - π

True / False

6. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
7. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
8. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** Because any permutation of the vertices of a large triangle can be obtained by rotation or reflection, the coloring of the large triangle is determined by which set of three colors is used for the corner triangles and the color that is used for the center triangle. If the three corner triangles are the same color, there are six possible sets of colors for them. If exactly two of the corner triangles are the same color, there are $6 \cdot 5 = 30$ possible sets of colors. If the three corner triangles are different colors, there are $\binom{6}{3} = 20$ possible sets of colors. Therefore there are $6 + 30 + 20 = 56$ sets of colors for the corner triangles. Because there are six choices for the color of the center triangle, there are $6 \cdot 56 = 336$ distinguishable triangles.

Note: Students should show all steps in their mathematical solution.

12. **Answer:** Subtracting the second column from the first column, we get

$$011x - 77 - 2x^3 - 34343 - 8 = 0.$$

The first column becomes all zeros when $x = 7$, so this is one possible value of x . Subtracting the third column from the first column, we get

$$011x + 27 - 2x^3 + 8343 - 8 = 0.$$

The first column becomes all zeros when $x = -2$, so this is another possible value of x . Expanding the last determinant along the first column, we get

$0 \begin{vmatrix} 1 & 1 \\ x+2 & 7 \end{vmatrix} + 1 \begin{vmatrix} x+2 & 7 \\ 7 & -2 \end{vmatrix} + 1 \begin{vmatrix} x^3+8 & 343 \\ 343 & -8 \end{vmatrix}$

$= -(x+2) \begin{vmatrix} 1 & 1 \\ 7 & -2 \end{vmatrix} + \begin{vmatrix} x^3+8 & 343 \\ 343 & -8 \end{vmatrix}$

$= -(x+2)(-9) + (x^3+8)(-351)$

$= 9(x+2) - 351(x+8)$

$= -9x^3 + 351x + 630 = 0$. Since we know that $x = 7$ and $x = -2$ are two solutions, we can take out a factor of $x - 7$ and $x + 2$, to get

$$-9(x - 7)(x + 2)(x + 5) = 0.$$

Hence, the possible values of x are 7, -2, and -5. (Alternatively, by Vieta's formulas, the sum of the three roots of $-9x^3 + 351x + 630 = 0$ is 0, so the third root is $-7 - (-2) = -5$.)

Note: Students should show all steps in their mathematical solution.

End of Answer Key